

Homework 1

1. A table of values of a function f is given below.

x	0	9	18	27	36
$f(x)$	20	15	-5	-10	5

- (a) Find a left-endpoint approximation with $n = 4$ for $\int_0^{36} f(x)dx$.

- (b) Find a right-endpoint approximation with $n = 4$ for $\int_0^{36} f(x)dx$.

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2. Suppose that

$$\int_4^8 f(x) \, dx = 1.7, \quad \int_6^8 f(x) \, dx = 2.5, \quad \text{and} \quad \int_4^8 (f(x))^2 \, dx = 5.$$

Evaluate each of the following integrals.

(a) $\int_4^8 (1 + f(x))^2 \, dx = \underline{\hspace{2cm}}.$

(b) $\int_4^6 f(x) \, dx = \underline{\hspace{2cm}}.$

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3. Use geometry and properties of integrals to evaluate

$$\int_{-3}^3 \left(|1 - 2x| + 9 + \sqrt{9 - x^2} \right) dx.$$

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4. Let $f(x) = x^2 - 2ax + 1$ where a is a positive constant. Find the value(s) of a such that the average value of f on the interval $[0, a]$ is 0.

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1. Use the Fundamental Theorem of Calculus to compute the following, without integrating anything.

(a) $\frac{d}{dx} \int_0^x \sqrt{1+t^2} dt$

(b) $\frac{d}{dy} \int_1^y 3x^2 dx$

(c) $\frac{d}{dz} \int_z^5 \sin(y^2) dy$

(d) $\frac{d}{dw} \int_w^{-2} \sec(z^3) dz$

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Problem 1 Continued.

(e) $\frac{d}{dv} \int_7^{v^2} \ln(w^2 + 1) \, dw$

(f) $\frac{d}{du} \int_3^{u^3+u} \tan(v) \, dv$

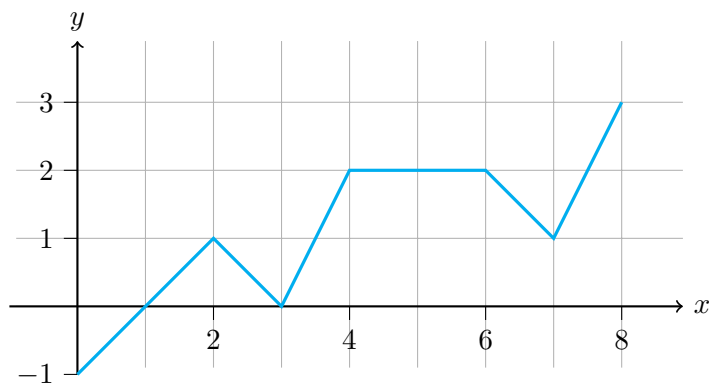
(g) $\frac{d}{ds} \int_{\sqrt{s}}^6 \frac{u^2}{u^2 + 4} \, du$

(h) $\frac{d}{dr} \int_{\cos(r)}^{\sin(r)} e^{s^2} \, ds$

(i) $\frac{d}{dt} \int_{\sqrt{t}}^{t^2} \sin^{-1}(r) \, dr$

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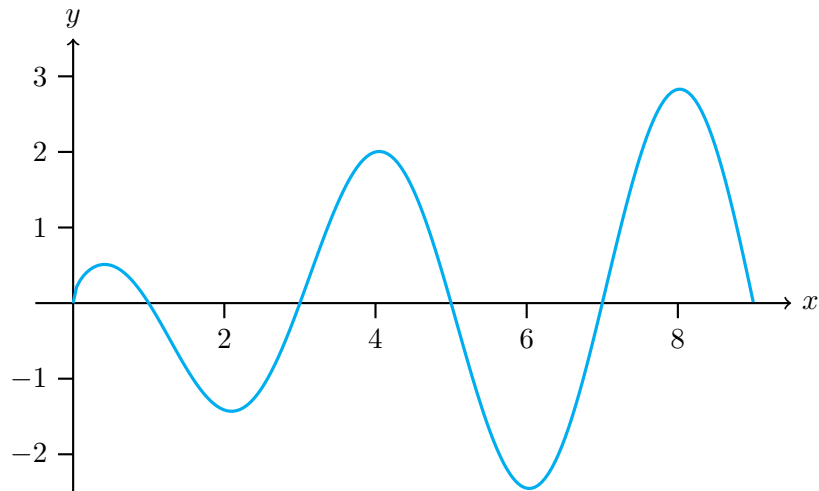
2. Find the average value of f , graphed below, on $[0, 8]$.



3. If $p(q)$ is an even function with $\int_0^3 p(q) dq = 6$, then the average value of $p(q)$ on the interval $[-3, 3]$ is _____.

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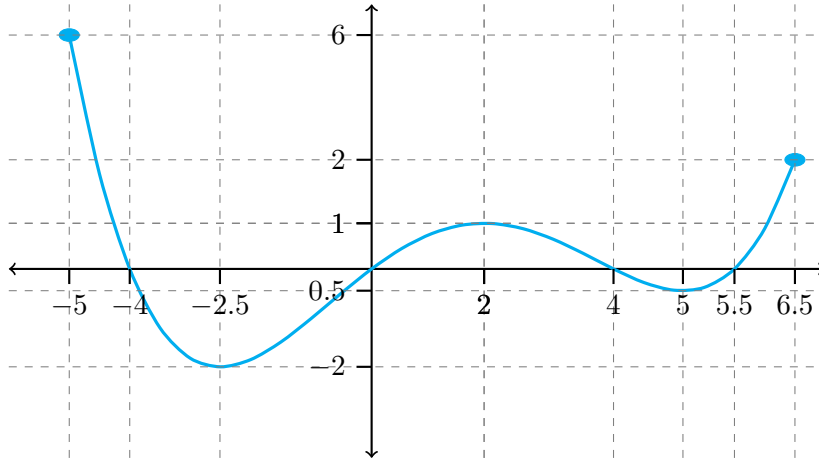
4. Let $g(x) = \int_0^x f(t) dt$, where f is the function whose graph is shown.



- (a) At what values of x do the local maximum and minimum values of g occur.
- (b) Where does g attain its absolute maximum value?
- (c) On what intervals is g concave downward?
- (d) Sketch the graph of g .

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5. The graph of $f(t)$, defined on the interval $[-5, 6.5]$, is given below. Define a function by $h(x) = \int_{-5}^x f(t) dt$ for $-5 \leq x \leq 6.5$.



- (a) Determine the interval(s) where $h(x)$ is increasing.
- (b) Determine the critical points of $h(x)$.
- (c) Find all local maximum points.
- (d) Determine the interval(s) where $h(x)$ is concave down.